**SkyStrat User Guide**

**Mapping Stratigraphic Height Using Structural Data and Digital Topography**

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## Overview

SkyStrat consists of two tools that work together to calculate and map stratigraphic height across a landscape. The first tool, the **Three-Point Problem Tool**, uses the elevations of three points on the same bedding surface to calculate the orientation of that surface. The second tool, the **Stratigraphic Height Mapping Tool** (the Busk method tool), takes those orientations together with a digital elevation model (DEM) and calculates the stratigraphic height of every point in your study area.

Used together, these tools allow you to answer the question: "How far up the sedimentary rock succession is any given point on the landscape?" — without measuring a section with a Jacob's staff.

Also see the instructional videos for the Three-Point Problem Tool and Stratigraphic Height Mapping Tool.

***Important note:*** *These tools will only function correctly if the map unit is the same unit as the elevation of the DEM. In most cases, where the DEM is in meters above sea level, this will require use of a projected coordinate system with meters as the unit. Users should take care that all input files and the GIS environment are in the correct coordinate system.*

Find the most recent version of the scripts at https://github.com/mitchellmcm27/SkyStrat

This guide was written for the ArcGIS Pro version of the scripts.

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## Part 1: The Three-Point Problem Tool

### What it does

Given three points that you identify as lying on the same bedding surface, this tool calculates the strike and dip of that surface. This is useful when visible beds can be traced across a DEM but cannot be directly measured in the field (for example, in remote or inaccessible areas, or when working from lidar data).

### What you need

- A point feature class containing three or more points, each lying on the same bedding surface. Points should be distributed across the surface rather than clustered together — wider spacing yields more accurate results. In addition, points should be distributed to form a triangular shape, as close to an equilateral triangle as possible. Co-linear points will result in unreliable strikes and dips.

- A second point feature class, which should start completely empty, which will have the strike/dip data from the spatial points stored in it. For reasons that have to do with the ArcPy environment, having the user generate and supply this feature class facilitates iterative additions of successive strike/dip measurements into the same file, rather than having Arc create a new strike/dip feature class for each run of the tool.
- A DEM covering the study area, from which the tool reads the elevation of each point.

**How to run it**

1. Load your point feature classes and DEM into ArcGIS Pro.
2. Open the Three-Point Problem Tool from the SkyStrat toolbox.
3. Select your point feature class, strike/dip feature class, and DEM.
4. Run the tool. It will append the strike and dip of the best-fit plane through your point triplets to the strike/dip tool you input.

**Tips**

- Points measured remotely from a high-resolution lidar DEM (1 meter or better) typically yield accurate results. Coarser DEMs introduce more uncertainty.
- Where possible, verify remotely measured orientations against field measurements.

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**Part 2: The Stratigraphic Height Mapping Tool**

**What it does**

This tool takes a set of strike and dip measurements distributed across a study area, together with a DEM and a polygon defining your study area, and calculates the stratigraphic height of every DEM grid cell within the study area. The result is a raster map where each cell's value is its height above (or below) a reference level in the stratigraphic succession.

**What you need**

- A **point feature class** containing your strike and dip measurements. Each point should have fields for strike (azimuth), dip (angle), and a unique label. An optional field can indicate whether beds are overturned at that location.
- A **DEM** covering the study area.
- A **polygon feature class** defining the study area boundary.
- You can force the algorithm to use a pre-defined **trend and plunge of the fold axis** for your study area. Otherwise, the algorithm will solve for the best fit fold axis and accompanying profile plane.
- The tool has space to specify a **stratigraphic height value** for the input measurement stations. If you choose to use this feature, only one of the input stations can have their stratigraphic height specified, and the other stations and DEM cells will be calculated relative to that stratigraphic height. If you do not use this feature, the right-most station on the profile graph will be set to have a stratigraphic height of zero and other stations and DEM cells will be calculated accordingly. If you want to recalculate the stratigraphic height map relative to a different datum later on, you can use the Raster Calculator to add a constant to the calculated map, which is mathematically equivalent to using a different specified datum at this step.

**Parameters**

| **Parameter** | **Description** |
|---|---|
| Strike and Dip Point Feature Class | Your input bedding orientation measurements |
| Strike Field | Field containing strike azimuths (degrees) |
| Dip Field | Field containing dip angles (degrees) |
| Overturned Flag Field (Optional) | Field with value 1 where beds are overturned, values of 0 or <Null> indicate upright |
| Label Field | A field with a unique value for each point (can be any unique identifier, including latitude or longitude; **using FID can result in errors**) |

| Parameter | Description |
|---|---|
| Stratigraphic Height Field (Optional) | Known stratigraphic heights at measurement points, if available (see discussion above) |
| Digital Elevation Model | Your input DEM raster |
| Study Area Polygon | Polygon defining the area to calculate heights for |
| Fold Axis Trend (Optional) | Azimuth of the fold axis (degrees); leave blank to calculate best fit |
| Fold Axis Plunge (Optional) | Plunge of the fold axis (degrees); leave blank to calculate best fit |
| Output PDF Path | Path for the down-plunge profile view figure |
| Plot DEM Cell Centroids on Profile | Check this to show DEM cells on the profile view (up to 10,000 cells are plotted) |
| Output Wedge Assignment Raster (Optional) | Raster showing which geometry region each DEM cell falls into |
| Output Strike/Dip Feature Class with Heights (Optional) | Copy of input points with calculated stratigraphic heights added |
| Output Stratigraphic Height Raster (Optional) | The main output, but technically optional: stratigraphic height mapped across the study area |

**How to run it**

1. Prepare your strike and dip feature class with the required fields.
2. Open the Stratigraphic Height Mapping Tool (Busk Method) from the SkyStrat toolbox.
3. Fill in the parameters as described above.
4. Run the tool. Tested computation times on a 2023 laptop were typically 10–20 minutes for a study area covering several square kilometers with a 1-meter DEM.

5. Check the output PDF for the down-plunge profile view — this is a useful diagnostic showing how your measurements project onto the profile plane and how the wedge or rectangle regions between them are defined.

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### How the Method Works

#### The profile plane and down-plunge projection

The tool begins by projecting all of your measurement points and DEM grid cells onto a single plane, the **profile plane**, which is oriented perpendicular to the fold axis. Viewing your data on this plane is equivalent to looking at the fold from directly down its axis (a "down-plunge view"). In this view, all beds appear as lines, and the geometry of the fold is most clearly visible. The up direction on the profile plane is the direction of the positive z vector from physical space projected onto the profile plane. The horizontal and vertical axes of the profile plane have the same scale.

#### Geometry between adjacent measurement points

Once the measurements are projected onto the profile plane, the tool processes each adjacent pair of measurements from left to right across the profile. For each pair, it determines the geometry of the region between them. There are two possible cases:

#### Case 1: Wedge (Circle Sector) Geometry

If the two adjacent bedding orientations are not parallel, lines drawn perpendicular to each bedding trace will intersect at some point. This intersection point is treated as the center of a set of concentric circles, and the arcs of these circles represent bedding surfaces. Any point in this region has a stratigraphic height that depends on its distance from the intersection point. This is the standard Busk method geometry and is the most common case.

#### Case 2: Rectangle Geometry

If two adjacent bedding orientations are parallel, their perpendicular lines will not intersect. In this case, the tool instead treats the region between the two measurements as a rectangle, with bedding surfaces running parallel to the measured orientations. Within a rectangle, stratigraphic height increases uniformly in the stratigraphic up direction perpendicular to bedding. This situation arises naturally where bedding dip is consistent across part of the study area — for instance, on the limb of a large fold far from the hinge.

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**When the tool cannot calculate stratigraphic heights**

The tool checks whether each pair of adjacent measurements produces a valid geometry before proceeding to calculate heights. There are situations where it cannot:

***Opposite stratigraphic directions between adjacent stations***

Each measurement has an associated "stratigraphic up" direction, the direction perpendicular to bedding pointing toward younger rocks, and an opposite "stratigraphic down" direction pointing toward older rocks.

When looking for the intersection point that defines the circle center for a wedge, the tool checks only whether the two stratigraphic-down vectors intersect (producing a **down-down wedge**, for an anticlinal geometry) or the two stratigraphic-up vectors intersect (producing an **up-up wedge**, for a synclinal geometry). If one measurement's stratigraphic-up vector would intersect with an adjacent measurement's stratigraphic-down vector, this means the two measurements are geometrically inconsistent with a smooth cylindrical fold generated using the Busk method. In this case the tool will warn you that no valid wedge geometry exists for that pair of measurements and will not calculate stratigraphic heights. The down-plunge profile view will still be generated, which can help you identify which measurements are causing the problem.

This scenario can arise from simple errors or genuine structural complexity, such as parasitic folding not captured by the existing set of strike/dip measurements or non-cylindrical folding. We suggest pursuing an iterative method of adding or removing additional measurements until the calculation runs and making sure to check the output by comparing contours of stratigraphic height to visible bedding features, etc., to check the accuracy of the result.

***Areas beyond the bounds of strike and dip measurements***

If any region of the study area falls outside all valid wedges and rectangles — for instance, at the edges of the study area beyond the outermost measurements — those DEM cells will not be assigned a stratigraphic height. Extending your measurement coverage will help.